PHT231 · Fall 2026

Pharmaceutical Dosage Forms 1

Question bank — 192 multiple-choice questions across 7 lectures. Answer key with explanations at the end.

Lecture 1 — Liquid dosage forms and solubilization

1. Syrup BP is an exception to the usual solute/solvent rule because it contains:
A. 33.3% w/w sucrose
B. 66.7% w/w sucrose
C. 50% w/v sucrose
D. 85% w/v sucrose

2. Paracetamol elixir is solubilized mainly by:
A. Hydrotropy
B. Micellization
C. Complexation
D. Co-solvency (alcohol + propylene glycol)

3. Which statement about hydrotropes is correct?
A. They have a CMC
B. Their hydrophobic part is too small to self-aggregate, so no CMC
C. They are cyclodextrins
D. They work at 0.05%

4. β-cyclodextrin is made of how many glucose units?
A. 6
B. 7
C. 8
D. 9

5. A BCS class II drug has:
A. High solubility, high permeability
B. Low solubility, high permeability
C. High solubility, low permeability
D. Low solubility, low permeability

6. A dosage form is defined as:
A. The active drug molecule
B. The physical form in which the drug is delivered to its site of action
C. The dose in mg
D. The route of administration

7. Why do solutions of aspirin or KCl cause less GIT irritation than solids?
A. They are diluted quickly by gastric contents
B. They contain buffers
C. They are absorbed in the mouth
D. They are not absorbed

8. Which is a disadvantage of oral solutions?
A. Rapid absorption
B. Uniform dose
C. Easy to swallow
D. Less stable due to hydrolysis

9. Solutions are especially suitable for which patients?
A. Athletes
B. Diabetics only
C. Only adults
D. Children and the elderly

10. About what fraction of lipophilic drug candidates fail because of poor bioavailability?
A. 4%
B. 40%
C. 80%
D. 100%

11. Bioavailability is:
A. Drug solubility in water
B. The dose per kg
C. The drug's half-life
D. The fraction of a dose reaching systemic circulation unchanged

12. BCS Class III drugs have:
A. High solubility, low permeability
B. Low solubility, high permeability
C. High solubility, high permeability
D. Low solubility, low permeability

13. Solubilization helps most for BCS class:
A. I
B. III
C. II
D. None

14. Why can salt formation reduce absorption of a BCS Class III drug?
A. It lowers solubility
B. It causes hydrolysis
C. The more polar, ionized form crosses membranes less
D. It increases particle size

15. Hydrocortisone sodium phosphate is an example of:
A. Co-solvency
B. Chemical modification
C. Hydrotropy
D. Complexation

16. Chloramphenicol sodium succinate:
A. Is inactive but converted to the active base in the body
B. Is the active antibacterial itself
C. Is insoluble
D. Is used orally only

17. Paracetamol elixir uses which solubilization method?
A. Micellization
B. Co-solvency (alcohol + propylene glycol)
C. Complexation
D. Hydrotropy

18. The CMC of most surfactants is about:
A. 0.0001%
B. 0.05–0.1%
C. 5–10%
D. 50%

19. Fat-soluble vitamins A, D, E, K are solubilized using:
A. Tweens (polysorbates)
B. Cyclodextrins
C. Urea
D. PEG 6000

20. Dettol and Lysol solubilize chloroxylenol and cresol with:
A. Sugar
B. Cyclodextrins
C. Ethanol only
D. Soaps

21. PEG–PLA is an example of:
A. An amphiphilic polymer forming polymeric micelles
B. A hydrotrope
C. A cyclodextrin
D. A preservative

22. Which statement about hydrotropes is TRUE?
A. They have a CMC
B. Their hydrophobic part is too small to self-aggregate
C. They are always polymers
D. They form inclusion complexes

23. Sodium benzoate, sodium salicylate and urea are examples of:
A. Co-solvents
B. Polymers
C. Surfactants
D. Hydrotropes

24. γ-Cyclodextrin contains how many glucose units?
A. 6
B. 7
C. 8
D. 9

25. Cyclodextrins are produced from:
A. Cellulose
B. Gelatin
C. Petroleum
D. Starch by enzymatic conversion

26. In a solid dispersion the drug is dispersed in:
A. An oil
B. Water only
C. A hydrophilic polymer matrix at molecular level
D. A cyclodextrin cavity

Lecture 2 — Particle size reduction: micronization, nanocrystals, cocrystals

27. According to Noyes–Whitney, dissolution rate is directly proportional to:
A. Particle size
B. Surface area
C. Diffusion layer thickness h
D. Drug concentration in the bulk C

28. Drug nanocrystals are in the size range:
A. 1–10 µm
B. 100–1000 nm
C. 1–10 nm
D. 10–100 µm

29. Why do nanocrystal formulations need stabilizers?
A. To raise the dose
B. Their high surface energy makes them aggregate
C. To mask taste
D. To lower pH

30. Entresto® (sacubitril + valsartan) is an example of a:
A. Nanocrystal
B. Multidrug cocrystal
C. Single-drug cocrystal
D. Solid dispersion

31. According to Noyes–Whitney, dissolution rate is directly proportional to:
A. Particle size
B. Surface area
C. Viscosity
D. Diffusion layer thickness

32. Micronization reduces particles to:
A. 1–10 nm
B. 100–1000 nm
C. 1–10 µm
D. 1–10 mm

33. Jet milling reduces particle size by:
A. Rotating balls
B. Precipitation
C. Particle-on-particle collisions in high-speed gas streams
D. Heat

34. Which drug is marketed in micronized form?
A. Griseofulvin
B. Sirolimus
C. Aprepitant
D. Fenofibrate (Tricor®)

35. Micronization may be insufficient for drugs with:
A. log P < 1
B. Low molecular weight
C. High solubility
D. log P > 4

36. Why may the FDA ask for evidence after micronization?
A. To show it changed the polymorph
B. To show dissolution improved without altering polymorphic form
C. To check color
D. To confirm sterility

37. Nanocrystals are in the range:
A. 1–10 nm
B. 100–1000 nm
C. 1–10 µm
D. 10–100 µm

38. High-pressure homogenization is a ___ method for nanocrystals.
A. Bottom-up
B. Chemical
C. Top-down
D. Cocrystal

39. A limitation of top-down nanocrystal production is:
A. Residual solvent
B. Uses no energy
C. Not scalable
D. Heat that can degrade thermolabile drugs

40. Solvent–antisolvent precipitation is a:
A. Top-down method
B. Cocrystal method
C. Micronization method
D. Bottom-up method

41. A drawback of bottom-up methods is:
A. Scale-up difficulty and residual solvent
B. Very high heat
C. Large particles
D. No control of shape

42. Nanocrystals need stabilizers because they:
A. Dissolve too slowly
B. Have high surface energy and aggregate
C. Are too large
D. Are toxic

43. SDS stabilizes nanocrystals mainly by:
A. Steric stabilization
B. Covalent bonding
C. Electrostatic repulsion (zeta potential)
D. Increasing size

44. PVP, HPMC and PEG stabilize nanocrystals by:
A. Steric
B. Electrostatic
C. Chemical reaction
D. Freezing

45. Which is a marketed nanocrystal product?
A. Entresto®
B. Emend® (aprepitant)
C. Griseofulvin
D. Calamine

46. In a cocrystal, drug and coformer are held by:
A. Covalent bonds
B. Ionic bonds only
C. Non-covalent interactions
D. Metallic bonds

47. Cocrystals change:
A. Pharmacological activity
B. Its mechanism of action
C. The drug molecule's structure
D. Physical properties only

48. Which is NOT a mechanism by which cocrystals improve solubility?
A. Forming covalent prodrugs
B. Improved wetting
C. Thermodynamic driving force
D. Crystal lattice modification

49. GRAS stands for:
A. Good Regulatory Approval Standard
B. Generally Recognized As Safe
C. General Rule of Active Substances
D. Granule Release Assay

50. Entresto® is a multidrug cocrystal of:
A. Meloxicam + aspirin
B. Caffeine + oxalic acid
C. Carbamazepine + saccharin
D. Sacubitril + valsartan

51. The meloxicam–aspirin cocrystal raised bioavailability from:
A. 16% to 69%
B. 6% to 9%
C. 50% to 60%
D. 1% to 100%

52. Caffeine–oxalic acid cocrystal illustrates improvement in:
A. Stability against humidity
B. Taste
C. Onset of action
D. Particle size

53. Liquid-assisted grinding differs from neat grinding by:
A. Dissolving fully
B. Using no liquid
C. Using microwaves
D. Using a little catalytic liquid for better efficiency and purity

Lecture 3 — Pharmaceutical waters and aromatic waters

54. Which water must be used for injections?
A. Purified water
B. Freshly boiled water
C. Potable water
D. Water for injection (pyrogen-free)

55. Water for injection is packed in single-dose containers of no more than:
A. 100 mL
B. 500 mL
C. 1 L
D. 5 L

56. In distillation of purified water, which portions are discarded?
A. First 10–20% and last 10–20%
B. Only the last 50%
C. Only the first 5%
D. None

57. Repeated redistillation of the distillate with fresh drug (e.g. rose water) is called:
A. Maceration
B. Percolation
C. Cohobation
D. Trituration

58. Tap water is not permitted for dispensing because of:
A. Taste only
B. Color
C. High pH only
D. Microbial contamination and dissolved salts

59. Freshly boiled and cooled water still contains:
A. Vegetative bacteria
B. Nothing
C. Pyrogens only
D. Spores and dissolved salts

60. Purified Water USP is about how many times freer of dissolved solids than ordinary water?
A. 2
B. 10
C. 100
D. 1,000

61. Purified Water USP must NOT be used for:
A. Oral solutions
B. USP assays
C. Injections and ophthalmic preparations
D. Syrups

62. In distillation, the first 10–20% of distillate is discarded because it contains:
A. Volatile impurities
B. Decomposition products
C. Pyrogens
D. Salts

63. Cation-exchange resin replaces Ca^{2+} and Mg^{2+} with:
A. OH^-
B. H^+
C. Na^+
D. Cl^-

64. An advantage of ion exchange over distillation is:
A. Removes pyrogens completely
B. Produces WFI
C. No heat needed
D. Needs complex apparatus

65. Reverse osmosis removes:
A. Only salts
B. Only microbes
C. All pyrogens, organics and 90–99% of ions
D. Nothing dissolved

66. Water for Injection is packaged in single-dose containers no larger than:
A. 100 mL
B. 500 mL
C. 1 L
D. 5 L

67. Bacteriostatic WFI is suitable for:
A. Large-volume infusions
B. Multiple-dose vials of small volume
C. Eye washes
D. Dialysis

68. Why must bacteriostatic WFI not be used in large volumes?
A. It is not sterile
B. Toxic amounts of preservative would be injected
C. It contains pyrogens
D. It is hypertonic

69. Aromatic waters are:
A. Concentrated oils
B. Alcoholic solutions
C. Nearly saturated clear aqueous solutions of volatile oils
D. Emulsions

70. Aromatic waters are incompatible with soluble salts because of:
A. Hydrolysis
B. Salting out
C. Oxidation
D. Complexation

71. Cohobation means:
A. Filtering through wetted paper
B. Adding Tween
C. Repeated redistillation until saturation
D. Boiling water

72. In the simple solution method, the oil–water mixture is set aside for:
A. 1 hour
B. 12 hours
C. 3 days
D. 1 week

73. Why is the filter paper wetted when preparing aromatic waters?
A. To speed filtration
B. So the paper doesn't absorb the oil
C. To sterilize
D. To add flavour

74. A disadvantage of using solubilizing agents in aromatic waters is:
A. They are too clear
B. Mould growth and bad taste above 2%
C. They are expensive
D. They evaporate

75. Which aromatic water is also used as a preservative?
A. Chloroform water
B. Anise water
C. Peppermint water
D. Camphor water

76. Camphor water is used in eye lotions as:
A. A preservative
B. A colouring
C. An antiseptic and anaesthetic
D. A buffer

Lecture 4 — Excipients in liquids, oral solutions and syrups

77. Borate buffers can be used:
A. For external application only
B. Parenterally
C. Orally
D. In all routes

78. Which antioxidant suits an oily (non-aqueous) preparation?
A. Ascorbic acid
B. Sodium citrate
C. BHA / BHT
D. EDTA only

79. Why do nearly saturated sucrose syrups not need a preservative?
A. Sucrose is antibacterial
B. They are sterile
C. There is no free water for microbes
D. They contain alcohol

80. A bitter drug is best masked with:
A. Anise, chocolate, mint or wild cherry
B. Apricot or peach
C. Citrus or raspberry
D. Vanilla only

81. Borate buffers are restricted to:
A. Oral use
B. Injections
C. External use
D. Eye drops only

82. Nasal drops should be buffered at about:
A. pH 2–3
B. pH 8–9
C. pH 5.3–7.0
D. pH 10

83. Benzylpenicillin injection is buffered to pH 6–7 with:
A. Sodium citrate
B. Borate
C. HCl
D. Ammonia

84. Which product must be isotonic?
A. Oral syrup
B. Eye drops
C. Liniment
D. Collodion

85. Potassium Citrate Mixture BP should be:
A. Taken undiluted
B. Applied to skin
C. Injected
D. Well diluted before taking (hypertonic)

86. Carbomer is used in ophthalmic preparations as a:
A. Preservative
B. Sweetener
C. Buffer
D. Viscosity enhancer

87. Which is NOT a source of contamination of solutions?
A. Equipment
B. Light-resistant containers
C. Personnel
D. Raw materials

88. Benzoic acid is used as a preservative at:
A. 0.001%
B. 0.1–0.2%
C. 5%
D. 20%

89. Propyl paraben differs from methyl paraben in being:
A. Water insoluble
B. Water soluble
C. Not a preservative
D. An antioxidant

90. Which antioxidant suits an aqueous preparation?
A. BHT
B. BHA
C. Ascorbic acid
D. Tocopherol

91. EDTA protects against oxidation by:
A. Absorbing light
B. Chelating trace heavy metals
C. Replacing oxygen
D. Lowering temperature

92. Replacing container headspace with nitrogen is done to:
A. Sterilize
B. Improve taste
C. Add pressure
D. Prevent oxidation

93. Sucrose is stable over a pH range of about:
A. 1–3
B. Any pH
C. 9–12
D. 4–8

94. Saccharin is used at concentrations below:
A. 0.2%
B. 2%
C. 20%
D. 50%

95. A bitter taste is best masked with:
A. Apricot
B. Vanilla only
C. Citrus
D. Anise, chocolate, mint or wild cherry

96. A salty taste is best masked with:
A. Wild cherry
B. Mint
C. Apricot, liquorice, peach, vanilla
D. Raspberry only

97. Menthol masks unpleasant tastes by:
A. A mild anaesthetic effect on taste receptors
B. Sweetening
C. Colouring
D. Precipitation

98. Which is a disadvantage of natural flavours?
A. Cheaper
B. More stable
C. Variable composition and microbial contamination
D. Always available

99. Colours are used in hospitals to:
A. Preserve
B. Thicken
C. Buffer
D. Differentiate antiseptic solutions

100. ORS contains per litre:
A. 45 mEq Na^+, 20 K^+, 35 Cl^-, 30 citrate, 25 g dextrose
B. 90 mEq Na^+ only
C. 10 g sucrose
D. 100 mEq K^+

101. Magnesium citrate oral solution is used as:
A. A saline cathartic
B. An antacid
C. A sedative
D. A diuretic

102. Sodium citrate–citric acid solution is used in gout because:
A. It dissolves uric acid crystals directly
B. It alkalinizes urine, preventing urate crystallization
C. It lowers blood pressure
D. It is a laxative

103. Sorbitol and glycerin syrups are NOT suitable for diabetics because they are:
A. Non-glycogenic
B. Insoluble
C. Toxic
D. Glycogenic

104. Artificial simple syrup contains:
A. Sucrose 66.7%
B. Sorbitol 70%
C. Sodium CMC 1.5% + saccharin sodium 0.6%
D. Glycerin 50%

105. Saccharin sodium is about how much sweeter than sucrose?
A. 3–5×
B. 30–55×
C. 300–550×
D. 3,000×

106. Nearly saturated sucrose syrups need no preservative because:
A. Sucrose is antibacterial
B. There is no free water for microbial growth
C. They are sterile
D. They are acidic

107. Alcohol does NOT act as a preservative below:
A. 1%
B. 5%
C. 15–20%
D. 50%

108. A linctus should be:
A. Diluted with water
B. Sipped slowly and not diluted
C. Injected
D. Applied externally

Lecture 5 — External aqueous and non-aqueous solutions

109. Retention enemas in adults should not exceed:
A. 50 mL
B. 150 mL
C. 500 mL
D. 1 L

110. Nasal decongestant solutions should be used for no more than 3–5 days to avoid:
A. Nose bleeds
B. Hypertension only
C. Loss of smell
D. Rhinitis medicamentosa

111. Flexible collodion contains, besides pyroxylin:
A. 2% camphor + 3% castor oil
B. 10% salicylic acid
C. Glycerin + alcohol
D. Starch

112. The FDA-proposed alcohol limit in OTC oral products for children under 6 is:
A. 0.5%
B. 5%
C. 10%
D. 20%

113. Lotions differ from liniments in that lotions are applied:
A. With rubbing
B. Only to broken skin
C. Without friction
D. Rectally

114. Glycerin in a lotion acts as a:
A. Preservative
B. Colour
C. Humectant
D. Propellant

115. Alcohol in a lotion gives:
A. Greasiness
B. A cooling effect and dry skin
C. Colour
D. Thickness

116. Which is an astringent in vaginal douches?
A. Povidone-iodine
B. Menthol
C. Potassium alum
D. Sodium lauryl sulfate

117. Sodium lauryl sulfate in douches acts as a:
A. Detergent
B. Buffer
C. Anaesthetic
D. Astringent

118. Plain water evacuation enemas use about:
A. 5–10 mL
B. 50 mL
C. 0.5–1 L
D. 5 L

119. Hydrocortisone retention enemas treat:
A. Constipation
B. Ulcerative colitis
C. Asthma
D. Insomnia

120. Aminophylline rectal enema is used for:
A. Asthma (systemic effect)
B. Sedation
C. Diagnosis
D. Evacuation

121. Diagnostic enemas contain:
A. Hydrocortisone
B. Glycerol
C. Barium sulfate
D. Phosphate

122. Retention enemas should not exceed about ___ in adults.
A. 15 mL
B. 150 mL
C. 500 mL
D. 1 L

123. Calcitonin nasal spray acts:
A. Locally as a decongestant
B. Systemically for osteoporosis
C. As an antiseptic
D. As a buffer

124. Nasal decongestant solutions are buffered at pH:
A. 2–3
B. 5.5–6.5
C. 8–9
D. 10–11

125. Rhinitis medicamentosa is caused by:
A. Prolonged decongestant use
B. Too-short use of decongestants
C. Isotonic drops
D. Antibiotics

126. Glycerin is used as an ear-drop vehicle because it is:
A. Volatile
B. Acidic
C. Viscous and hygroscopic
D. Antibacterial by itself only

127. Benzocaine in ear drops acts as:
A. Local anaesthetic/analgesic
B. Anti-infective
C. Anti-inflammatory
D. Wax remover

128. Oily IM depot injections of testosterone esters work by:
A. Rapid absorption
B. Hydrolysis in the oil
C. Dissolving in blood immediately
D. Slow partitioning of drug from the oil pool

129. Ethanol is useful for extracting crude drugs because it dissolves:
A. Gums and starch
B. Only water-soluble materials
C. Resins, volatile oils, alkaloids and glycosides, but not gums or starch
D. Proteins only

130. The OTC oral alcohol limit for children under 6 is:
A. 0.5%
B. 5%
C. 10%
D. 15%

131. Adding water to a tincture may cause:
A. Sterilization
B. Colour change only
C. Precipitation of drug
D. Gas

132. Alcohol in elixirs precipitates:
A. Glucose
B. Water
C. Tragacanth, acacia, agar and inorganic salts
D. Volatile oils

133. A disadvantage of glycerin as a solvent is that it:
A. Is toxic
B. Is volatile
C. Is not selective and very viscous
D. Is insoluble in water

134. Almond oil is used for phenol injections because:
A. Phenol is insoluble in oil
B. Aqueous phenol is caustic
C. It is cheaper
D. It is a preservative

135. Which oil is used as a solvent for oral antibiotics?
A. Castor oil
B. Coconut oil
C. Almond oil
D. Sesame oil

136. Alcoholic liniments are used for their:
A. Massage only
B. Nutrition
C. Sterility
D. Rubefacient and counter-irritant effects

137. Liniments should never be applied to:
A. Intact skin
B. Broken or injured skin
C. Muscles
D. Joints

138. Collodion contains pyroxylin at:
A. 0.4%
B. 4%
C. 40%
D. 10%

139. In flexible collodion, camphor's role is to:
A. Make it waterproof
B. Add flexibility
C. Act as solvent
D. Kill bacteria

140. Salicylic acid collodion contains salicylic acid at:
A. 1%
B. 10%
C. 25%
D. 50%

Lecture 6 — Colloids: classes, preparation and purification

141. Colloidal particles are in the size range:
A. < 1 nm
B. 1–500 nm
C. > 500 nm
D. 1–10 µm

142. Lyophobic sols are unstable in the presence of small amounts of electrolytes because:
A. They lose their solvent sheath
B. They form micelles
C. The charge on their particles is neutralized
D. They become viscous

143. Dialysis separates colloids from impurities based on:
A. Particle size, using a semipermeable membrane
B. Charge
C. Density
D. Colour

144. Colloidal particles range in size from:
A. <1 nm
B. 1–500 nm
C. 0.5–5 mm
D. >1 mm

145. Colloidal particles pass through:
A. Semipermeable membranes
B. Nothing
C. Filter paper but not semipermeable membranes
D. Ultrafilters

146. Colloidal particles are detected by the:
A. Naked eye
B. Light microscope
C. Ultramicroscope
D. None

147. Which is a molecular dispersion?
A. Suspension
B. Glucose solution
C. Emulsion
D. Colloidal silver

148. Platinum black is an effective catalyst because of its:
A. Colour
B. Huge specific surface area
C. Charge
D. Weight

149. When a red gold sol's particles grow larger, its colour becomes:
A. Green
B. Blue
C. Yellow
D. Colourless

150. Colloidal silver salts are preferred as germicides because they:
A. Don't irritate like ionic silver salts
B. Are cheaper
C. Are coloured
D. Dissolve fast

151. Dextran, PVP and gelatin are used as:
A. Plasma substitutes
B. Antacids
C. Catalysts
D. Germicides

152. Iron–dextran is used to treat:
A. Syphilis
B. Infections
C. Cancer
D. Anaemia

153. Lyophilic colloids form:
A. Spontaneously
B. Only by special methods
C. Only with electrolytes
D. Only by ultrasound

154. Hydration refers to:
A. Solvation when water is the medium
B. Loss of charge
C. Precipitation
D. Micelle formation

155. Rubber in an organic solvent is an example of a:
A. Hydrophilic colloid
B. Association colloid
C. Lyophobic colloid
D. Lipophilic (lyophilic) colloid

156. Lyophobic colloids are stabilized by:
A. Solvent sheath
B. Both sheath and charge
C. Surface charge only
D. Nothing

157. Lyophobic colloids affect the medium's viscosity:
A. Greatly
B. Only when heated
C. Hardly at all
D. By solidifying it

158. Lyophilic colloids are salted out by:
A. High concentrations of very soluble electrolytes (desolvation)
B. Small amounts of electrolyte
C. Heat
D. Light

159. In association colloids, electrolytes:
A. Raise the CMC
B. Have no effect
C. Lower the CMC
D. Destroy micelles always

160. The 'hedgehog' analogy says a particle in an unsuitable solvent:
A. Unrolls
B. Gains charge
C. Dissolves
D. Rolls up to reduce exposed area

161. An ultrasonic generator for dispersion operates above:
A. 200 cycles/s
B. 2,000 cycles/s
C. 20,000 cycles/s
D. 2 million cycles/s

162. A disadvantage of the colloid mill is:
A. Very high efficiency
B. It uses electricity
C. Low efficiency, reducing only a small proportion of particles
D. It forms nuclei

163. Pouring sulfur dissolved in an organic solvent into excess water makes a sol by:
A. Dispersion
B. Hydrolysis
C. Dialysis
D. Condensation (supersaturation and nucleation)

164. Boiling ferric chloride with water forms a red ferric oxide sol by:
A. Hydrolysis
B. Oxidation
C. Reduction
D. Milling

165. Why purify lyophobic sols?
A. To add colour
B. To increase viscosity
C. Electrolyte impurities cause flocculation
D. To sterilize

166. Electrodialysis uses a vessel with how many compartments?
A. Three
B. Two
C. One
D. Five

167. In electrodialysis, cations migrate toward the:
A. Positive electrode
B. Negative electrode
C. Sol compartment
D. They don't move

168. Ultrafiltration membranes are supported by wire screens because:
A. They are coloured
B. They are magnetic
C. They dissolve
D. High pressure or suction is applied

169. In haemodialysis, which substance moves from dialysate INTO blood?
A. Urea
B. Plasma proteins
C. Glucose
D. Bicarbonate

170. Glucose in dialysate is kept at the same concentration as blood so that:
A. It diffuses out
B. There is no net loss or gain
C. It precipitates
D. It kills bacteria

171. Success of the artificial kidney is judged by its ability to reduce blood:
A. Glucose
B. Albumin
C. Urea
D. Bicarbonate

Lecture 7 — Properties of colloids

172. The visible cone of light through a colloidal dispersion is the:
A. Brownian effect
B. Doppler effect
C. Donnan effect
D. Faraday–Tyndall effect

173. Brownian motion increases when:
A. Particle size and viscosity increase
B. The particles are > 5 µm
C. Glycerin is added
D. Particle size and viscosity decrease

174. By Stokes' law, doubling the particle radius makes sedimentation velocity:
A. Double
B. Half
C. Four times faster
D. Unchanged

175. The Tyndall cone results from:
A. Absorption of light
B. Fluorescence
C. Scattering of light by colloidal particles
D. Reflection by the container

176. Turbidity of a colloid is measured at:
A. 0°
B. 45°
C. 90°
D. 180°

177. Turbidity of a lyophobic colloid is proportional to:
A. Viscosity
B. Molecular weight and concentration
C. Temperature only
D. pH

178. In the ultramicroscope, colloidal particles appear as:
A. Dark spots on light background
B. Bright spots on dark background
C. Their true shape
D. Coloured rings

179. The main drawback of the electron microscope for colloids is:
A. Low resolution
B. Only dried samples can be examined (no solvation info)
C. It can't show shape
D. It's too fast

180. Resolving power is:
A. Magnification
B. Light intensity
C. The smallest distance at which two objects remain distinguishable
D. Wavelength

181. Brownian motion is caused by:
A. Gravity
B. Electric fields
C. Bombardment by molecules of the dispersion medium
D. Light

182. Brownian motion is faster with:
A. Larger particles
B. Smaller particles and lower viscosity
C. Higher viscosity
D. Lower temperature

183. Diffusion of colloids is a direct result of:
A. Sedimentation
B. Viscosity
C. Osmosis
D. Brownian motion

184. The minus sign in Fick's first law indicates:
A. Diffusion goes toward decreasing concentration
B. Diffusion is negative
C. Particles are negative
D. Time is reversed

185. Osmosis is the movement of:
A. Solute from high to low concentration
B. Colloids through filter paper
C. Solvent from lower to higher solute concentration across a semipermeable membrane
D. Ions in an electric field

186. In Stokes' law, halving the particle radius changes settling velocity to:
A. A quarter
B. Half
C. Double
D. Unchanged

187. Increasing medium viscosity in Stokes' law:
A. Decreases settling
B. Increases settling
C. Has no effect
D. Reverses settling

188. Below about 0.5 µm, sedimentation is opposed by:
A. Gravity
B. Viscosity alone
C. Brownian motion
D. Charge

189. The ultracentrifuge is used to determine:
A. Molecular weight of proteins and sample homogeneity
B. Colour
C. pH
D. Turbidity

190. In $\eta(int) = K \cdot M^a$, K and a are:
A. Universal constants
B. Variables
C. Constants characteristic of the polymer–solvent system
D. Temperature terms

191. Linear lyophilic particles give high viscosity because they are:
A. Unsolvated and spherical
B. Solvated and asymmetric
C. Charged only
D. Small

192. Electrophoresis is the:
A. Scattering of light
B. Settling under gravity
C. Movement of charged colloidal particles in an electric field
D. Diffusion

Answer key

Lecture 1

1	**B**	**66.7% w/w sucrose** — The sucrose (solute) is the larger part: 66.7% w/w in 33.3% w/w water.
2	**D**	**Co-solvency (alcohol + propylene glycol)** — Water-miscible co-solvents in which paracetamol is soluble.
3	**B**	**Their hydrophobic part is too small to self-aggregate, so no CMC** — Unlike surfactants, hydrotropes (e.g. sodium benzoate, urea) don't form micelles.
4	**B**	**7** — α = 6, β = 7, γ = 8.
5	**B**	**Low solubility, high permeability** — Dissolution limits absorption, so solubilization helps these most.
6	**B**	**The physical form in which the drug is delivered to its site of action** — Tablets, syrups, injections are dosage forms.
7	**A**	**They are diluted quickly by gastric contents** — No concentrated drug sits on the mucosa.
8	**D**	**Less stable due to hydrolysis** — Water promotes hydrolysis; also bulky, microbial growth, taste.
9	**D**	**Children and the elderly** — They are easier to swallow.
10	**B**	**40%** — About 40%.
11	**D**	**The fraction of a dose reaching systemic circulation unchanged** — Poor solubility often means poor bioavailability.
12	**A**	**High solubility, low permeability** — Permeation is limiting for Class III.
13	**C**	**II** — Class II is limited by dissolution only.
14	**C**	**The more polar, ionized form crosses membranes less** — If permeability is limiting, more polarity hurts.
15	**B**	**Chemical modification** — A water-soluble derivative of the drug.
16	**A**	**Is inactive but converted to the active base in the body** — It is a water-soluble form suitable for parenteral use.
17	**B**	**Co-solvency (alcohol + propylene glycol)** — Water-miscible co-solvents raise solubility.
18	**B**	**0.05–0.1%** — Above it, micelles form.
19	**A**	**Tweens (polysorbates)** — Micellar solubilization.
20	**D**	**Soaps** — Soap micelles take up the phenols.
21	**A**	**An amphiphilic polymer forming polymeric micelles** — Hydrophobic core traps drug; hydrophilic shell stabilizes.
22	**B**	**Their hydrophobic part is too small to self-aggregate** — No micelles, no critical concentration.
23	**D**	**Hydrotropes** — Used in large amounts to raise solubility.
24	**C**	**8** — α = 6, β = 7, γ = 8.
25	**D**	**Starch by enzymatic conversion** — Enzymatic conversion of starch.
26	**C**	**A hydrophilic polymer matrix at molecular level** — PVP, HPMC or PEG 6000, by melting or kneading.

Lecture 2

27	**B**	**Surface area** — dm/dt = D·A·(Cs − C)/h; smaller particles → larger A → faster dissolution.
28	**B**	**100–1000 nm** — Micronized = 1–10 µm; nanocrystals = 100–1000 nm.
29	**B**	**Their high surface energy makes them aggregate** — Polymers (steric) or charged surfactants (electrostatic) keep particles apart.
30	**B**	**Multidrug cocrystal** — Two APIs in one crystal lattice, held by non-covalent bonds.
31	**B**	**Surface area** — dm/dt = D·A·(Cs−C)/h.
32	**C**	**1–10 µm** — Micrometre range, mechanically.
33	**C**	**Particle-on-particle collisions in high-speed gas streams** — Compressed air or nitrogen streams.
34	**A**	**Griseofulvin** — Also spironolactone.
35	**D**	**log P > 4** — Very lipophilic drugs need more than micronization.
36	**B**	**To show dissolution improved without altering polymorphic form** — Milling energy can convert crystal forms.
37	**B**	**100–1000 nm** — Below one micrometre.
38	**C**	**Top-down** — It breaks coarse crystals down.

39	D	**Heat that can degrade thermolabile drugs** — Also high energy consumption.
40	D	**Bottom-up method** — It builds particles from dissolved molecules.
41	A	**Scale-up difficulty and residual solvent** — Also more complex process control.
42	B	**Have high surface energy and aggregate** — Stabilizers keep particles apart.
43	C	**Electrostatic repulsion (zeta potential)** — Charged stabilizers create repulsion.
44	A	**Steric** — Polymer coating prevents close contact.
45	B	**Emend® (aprepitant)** — Also Rapamune® and Tricor®.
46	C	**Non-covalent interactions** — H-bonds, van der Waals, π–π.
47	D	**Physical properties only** — Solubility, stability, strength — not pharmacology.
48	A	**Forming covalent prodrugs** — No covalent change occurs.
49	B	**Generally Recognized As Safe** — An FDA safety status required for coformers.
50	D	**Sacubitril + valsartan** — Used for symptomatic heart failure.
51	A	**16% to 69%** — A striking multidrug-cocrystal example.
52	A	**Stability against humidity** — Less hygroscopic than pure caffeine.
53	D	**Using a little catalytic liquid for better efficiency and purity** — Neat grinding is solvent-free.

Lecture 3

54	D	**Water for injection (pyrogen-free)** — Purified water is not allowed for parenteral or ophthalmic products.
55	C	**1 L** — ≤ 1 L and with no antimicrobial agent.
56	A	**First 10–20% and last 10–20%** — The first part carries volatile impurities; the last carries decomposition products.
57	C	**Cohobation** — Cohobation is used when the volatile oil content is small.
58	D	**Microbial contamination and dissolved salts** — Salts can cause precipitation, discoloration, effervescence.
59	D	**Spores and dissolved salts** — Boiling kills vegetative cells only.
60	C	**100** — About 100 times.
61	C	**Injections and ophthalmic preparations** — It isn't required to be sterile or pyrogen-free.
62	A	**Volatile impurities** — The last 10–20% is discarded for decomposition products.
63	B	**H^+** — Anion resin swaps anions for OH^-; $H^+ + OH^- \rightarrow H_2O$.
64	C	**No heat needed** — Also cheap, simple and mobile.
65	C	**All pyrogens, organics and 90–99% of ions** — Pressure forces water through a semipermeable membrane.
66	C	**1 L** — And contains no antimicrobial.
67	B	**Multiple-dose vials of small volume** — If more than 5 mL is needed, use sterile WFI.
68	B	**Toxic amounts of preservative would be injected** — Preservative toxicity limits volume.
69	C	**Nearly saturated clear aqueous solutions of volatile oils** — Used as pleasant-flavoured vehicles.
70	B	**Salting out** — Salts reduce oil solubility.
71	C	**Repeated redistillation until saturation** — Used for rose water.
72	B	**12 hours** — 2 g or 2 mL oil + 1000 mL water, 12 h, then filter.
73	B	**So the paper doesn't absorb the oil** — A dry paper would retain oil.
74	B	**Mould growth and bad taste above 2%** — E.g. Tween 20, Span 60.
75	A	**Chloroform water** — Also expectorant, carminative, antiseptic, stomachic.
76	C	**An antiseptic and anaesthetic** — Internally it is carminative.

Lecture 4

77	A	**For external application only** — Borates are for skin lotions, mouthwash, eyewash, not oral or parenteral use.
78	C	**BHA / BHT** — Tocopherol, BHA and BHT for oils; ascorbic acid for aqueous systems.
79	C	**There is no free water for microbes** — Microbes need free water; saturated syrup is effectively anhydrous for them.

80	A	**Anise, chocolate, mint or wild cherry** — Salty → apricot/peach; sour → citrus/raspberry; bitter → anise, chocolate, mint, wild cherry.
81	C	**External use** — Never oral or parenteral.
82	C	**pH 5.3–7.0** — Mean 6.3, matching the nasal cavity.
83	A	**Sodium citrate** — To slow autocatalytic acid hydrolysis.
84	B	**Eye drops** — Injections, eye and nasal preparations.
85	D	**Well diluted before taking (hypertonic)** — It is hypertonic.
86	D	**Viscosity enhancer** — Increases retention time in the eye.
87	B	**Light-resistant containers** — The three sources are equipment, raw materials and personnel.
88	B	**0.1–0.2%** — Parabens 0.1%; alcohol >20%.
89	A	**Water insoluble** — Methyl paraben is water soluble.
90	C	**Ascorbic acid** — BHA, BHT, tocopherol are for oils.
91	B	**Chelating trace heavy metals** — Metals catalyse oxidation.
92	D	**Prevent oxidation** — Removes oxygen.
93	D	**4–8** — Also colourless and very soluble.
94	A	**0.2%** — Higher amounts give bitter/metallic aftertaste.
95	D	**Anise, chocolate, mint or wild cherry** — Salty → apricot, liquorice, peach, vanilla.
96	C	**Apricot, liquorice, peach, vanilla** — Sour → citrus, liquorice, raspberry.
97	A	**A mild anaesthetic effect on taste receptors** — It desensitizes taste buds.
98	C	**Variable composition and microbial contamination** — Artificial flavours are cheaper and more stable.
99	D	**Differentiate antiseptic solutions** — Also identify poisons and match flavours.
100	A	**45 mEq Na^+, 20 K^+, 35 Cl^-, 30 citrate, 25 g dextrose** — For mild to moderate dehydration.
101	A	**A saline cathartic** — Usual adult dose 200 mL.
102	B	**It alkalinizes urine, preventing urate crystallization** — Urates crystallize in acid urine.
103	D	**Glycogenic** — They are converted to glucose.
104	C	**Sodium CMC 1.5% + saccharin sodium 0.6%** — For diabetic patients.
105	C	**300–550×** — 300–550 times, with a bitter taste.
106	B	**There is no free water for microbial growth** — Water is bound to sugar.
107	C	**15–20%** — Concentrations under 15–20% don't preserve.
108	B	**Sipped slowly and not diluted** — Demulcent action on the throat.

Lecture 5

109	B	**150 mL** — Larger volumes stimulate evacuation, so the drug would not be retained.
110	D	**Rhinitis medicamentosa** — Rebound congestion (rhinitis medicamentosa) worsens the original symptom.
111	A	**2% camphor + 3% castor oil** — Castor oil adds flexibility; camphor makes it waterproof.
112	A	**0.5%** — Under 6: 0.5%; 6–12: 5%; over 12 and adults: 10%.
113	C	**Without friction** — Liniments are rubbed in.
114	C	**Humectant** — Retains moisture on the skin.
115	B	**A cooling effect and dry skin** — It evaporates quickly.
116	C	**Potassium alum** — Also ammonium alum.
117	A	**Detergent** — Detergent component.
118	C	**0.5–1 L** — Small-volume ones (~130 mL) use $MgSO_4$ or phosphate.
119	B	**Ulcerative colitis** — Local anti-inflammatory effect.
120	A	**Asthma (systemic effect)** — Chloral hydrate enema is sedative.
121	C	**Barium sulfate** — A radiopaque agent for X-ray of the colon.
122	B	**150 mL** — Larger volumes trigger expulsion.
123	B	**Systemically for osteoporosis** — Ephedrine is the local decongestant example.
124	B	**5.5–6.5** — To protect cilia and drug stability.

125	A	**Prolonged decongestant use** — Limit use to 3–5 days.
126	C	**Viscous and hygroscopic** — Longer contact, less microbial growth, less swelling.
127	A	**Local anaesthetic/analgesic** — Often with antipyrine.
128	D	**Slow partitioning of drug from the oil pool** — The oil remains as a discrete entity in muscle.
129	C	**Resins, volatile oils, alkaloids and glycosides, but not gums or starch** — It is selective for actives.
130	A	**0.5%** — 6–12 years: 5%; over 12: 10%.
131	C	**Precipitation of drug** — Diluting the alcohol reduces solubility.
132	C	**Tragacanth, acacia, agar and inorganic salts** — A classic incompatibility.
133	C	**Is not selective and very viscous** — It dissolves gums and starch too.
134	B	**Aqueous phenol is caustic** — Used to block nerves and for haemorrhoids.
135	B	**Coconut oil** — Castor, sesame, olive oils are for parenteral depots.
136	D	**Rubefacient and counter-irritant effects** — Oily liniments are milder, for massage.
137	B	**Broken or injured skin** — Safety rule.
138	B	**4%** — In alcohol/ether or alcohol/acetone 1:3.
139	A	**Make it waterproof** — Castor oil 3% adds flexibility.
140	B	**10%** — Keratolytic for corns and warts.

Lecture 6

141	B	**1–500 nm** — They pass filter paper but not semipermeable membranes.
142	C	**The charge on their particles is neutralized** — They depend on surface charge only; lyophilic colloids also have a solvent sheath.
143	A	**Particle size, using a semipermeable membrane** — Small ions and molecules pass the membrane; colloidal particles stay.
144	B	**1–500 nm** — Molecular <1 nm; coarse >500 nm.
145	C	**Filter paper but not semipermeable membranes** — The basis of dialysis.
146	C	**Ultramicroscope** — Electron microscope shows them too.
147	B	**Glucose solution** — True solution < 1 nm.
148	B	**Huge specific surface area** — Catalysts adsorb reactants on their surface.
149	B	**Blue** — Colour relates to particle size.
150	A	**Don't irritate like ionic silver salts** — Silver iodide, chloride, protein.
151	A	**Plasma substitutes** — Hydrophilic colloids restoring blood volume.
152	D	**Anaemia** — Non-ionic hydrophilic sol.
153	A	**Spontaneously** — Strong attraction to the medium.
154	A	**Solvation when water is the medium** — The solvent sheath in water.
155	D	**Lipophilic (lyophilic) colloid** — Lyophilic in an organic medium.
156	C	**Surface charge only** — Small electrolyte amounts neutralize the charge.
157	C	**Hardly at all** — Unsolvated, symmetric particles.
158	A	**High concentrations of very soluble electrolytes (desolvation)** — They resist small amounts.
159	C	**Lower the CMC** — Salting out at higher concentration.
160	D	**Rolls up to reduce exposed area** — Lyophobic behaviour.
161	C	**20,000 cycles/s** — High-intensity ultrasound.
162	C	**Low efficiency, reducing only a small proportion of particles** — Material is sheared between rotating plates.
163	D	**Condensation (supersaturation and nucleation)** — Small nuclei grow to colloidal size.
164	A	**Hydrolysis** — A chemical condensation method.
165	C	**Electrolyte impurities cause flocculation** — Dialysis, electrodialysis, ultrafiltration.
166	A	**Three** — Electrodes in the outer two, sol in the middle.
167	B	**Negative electrode** — Anions move to the positive electrode.
168	D	**High pressure or suction is applied** — Very fine pores need support.

169	D	**Bicarbonate** — Deficient substances diffuse into blood.
170	B	**There is no net loss or gain** — Normal-level substances stay unchanged.
171	C	**Urea** — Urea diffuses out into dialysate.

Lecture 7

172	D	**Faraday–Tyndall effect** — Colloidal particles scatter light; true solutions appear clear.
173	D	**Particle size and viscosity decrease** — Smaller particles in a less viscous medium move faster.
174	C	**Four times faster** — $v \propto r^2$, so $2^2 = 4$ times faster.
175	C	**Scattering of light by colloidal particles** — True solutions don't scatter.
176	C	**90°** — At 90° to the incident beam.
177	B	**Molecular weight and concentration** — Used for molecular weight determination.
178	B	**Bright spots on dark background** — Based on light scattering; useful for counting.
179	B	**Only dried samples can be examined (no solvation info)** — The solvent sheath is lost on drying.
180	C	**The smallest distance at which two objects remain distinguishable** — Electron microscopes resolve down to ~5 Å.
181	C	**Bombardment by molecules of the dispersion medium** — Random zig-zag movement.
182	B	**Smaller particles and lower viscosity** — Glycerin slows and stops it.
183	D	**Brownian motion** — Described by Fick's first law.
184	A	**Diffusion goes toward decreasing concentration** — $dm/dt = -D \cdot A \cdot dC/dx$.
185	C	**Solvent from lower to higher solute concentration across a semipermeable membrane** — The pressure involved is osmotic pressure.
186	A	**A quarter** — $v \propto r^2$.
187	A	**Decreases settling** — η is in the denominator.
188	C	**Brownian motion** — An ultracentrifuge is needed.
189	A	**Molecular weight of proteins and sample homogeneity** — Also separates polydisperse systems.
190	C	**Constants characteristic of the polymer–solvent system** — Used for molecular weight from viscosity.
191	B	**Solvated and asymmetric** — Spherical lyophobic particles give low viscosity.
192	C	**Movement of charged colloidal particles in an electric field** — Linked to the electric double layer.